

Question 1 – Bank Management System

Date: 24/07/2026

Objective / Aim

To design and implement a Bank Management System in Java demonstrating inheritance, method overriding, deposits, withdrawals, interest calculation, fund transfer, and account details.

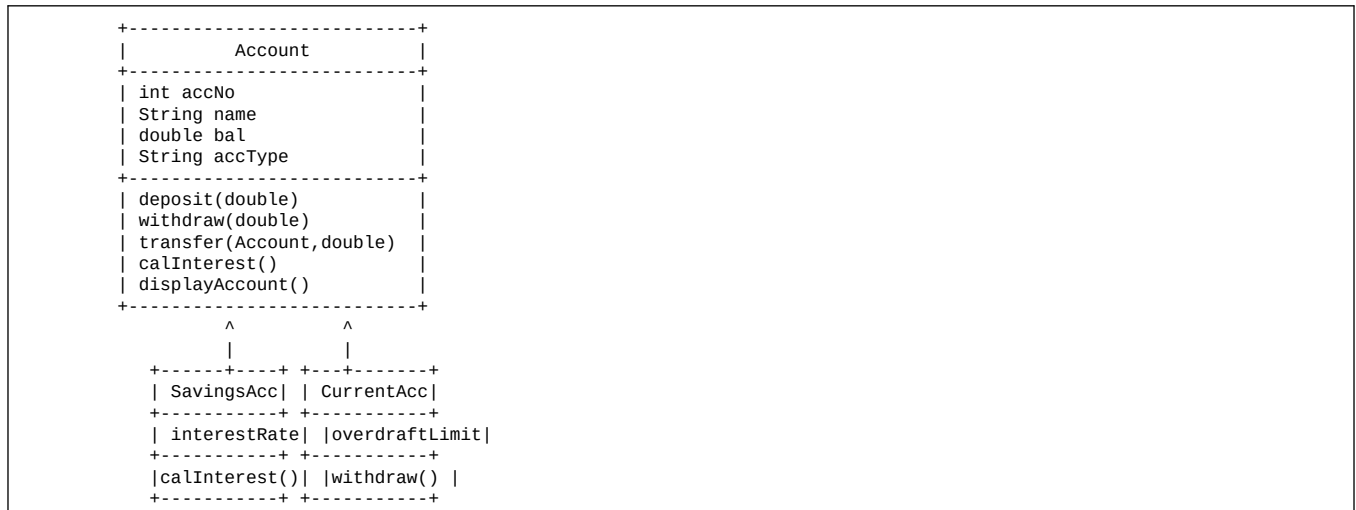
Algorithm / Logic

1. Create the Account base class with account number, name, balance, and account type.
2. Define deposit(), withdraw(), transfer(), callInterest(), and displayAccount() methods.
3. Create SavingsAcc extending Account and add interestRate.
4. Override callInterest() in SavingsAcc.
5. Create CurrentAcc extending Account and add overdraftLimit.
6. Override withdraw() in CurrentAcc to allow withdrawals up to the overdraft limit.
7. Create SavingsAcc and CurrentAcc objects and provide a menu-driven system.
8. Perform deposit, withdrawal, transfer, interest calculation, and display operations.

Steps for Execution

1. Create the Java project and add Account, SavingsAcc, CurrentAcc, and Bank.
2. Compile and run Bank.java.
3. Choose an operation from the bank menu.
4. Select the required account and enter the amount when requested.
5. Observe the transaction result.
6. Use Display Accounts to view account details.
7. Choose Exit to terminate the program.

UML Class Diagram



Source Code

```
import java.util.Scanner;
class Account {
    int accNo;
    String name;
    double bal;
    String accType;
    public Account(int accNo, String name, double bal, String accType) {
        this.accNo = accNo;
        this.name = name;
        this.bal = bal;
        this.accType = accType;
    }
    public void deposit(double amount) {
        if (amount > 0) {
            bal += amount;
            System.out.println("■" + amount + " deposited successfully.");
        } else {
            System.out.println("Invalid deposit amount.");
        }
    }
    public void withdraw(double amount) {
        if (amount <= bal) {
            bal -= amount;
            System.out.println("■" + amount + " withdrawn successfully.");
        } else {
            System.out.println("Insufficient Balance!");
        }
    }
    public void transfer(Account rec, double amount) {
        if (amount <= bal) {
            bal -= amount;
            rec.bal += amount;
            System.out.println("Transfer of " + amount + " to " + rec.name + " successful.");
        } else {
            System.out.println("Insufficient Balance!");
        }
    }
    public void callInterest() {
        // Interest calculation logic
    }
    public void displayAccount() {
        System.out.println("Account Details: " + name + ", " + accNo + ", " + bal + ", " + accType);
    }
}
```

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        rec.bal += amount;
        System.out.println("■" + amount + " transferred to " + rec.name);
    } else {
        System.out.println("Transfer Failed! Insufficient Balance.");
    }
}

public void calInterest() {
    System.out.println("Interest calculation not applicable.");
}

public void displayAccount() {
    System.out.println("\n-----");
    System.out.println("Account Number : " + accNo);
    System.out.println("Account Holder : " + name);
    System.out.println("Account Type : " + accType);
    System.out.println("Balance : ■" + bal);
}
}

class SavingsAcc extends Account {
    double interestRate;
    public SavingsAcc(int accNo, String name, double bal, double interestRate) {
        super(accNo, name, bal, "Savings");
        this.interestRate = interestRate;
    }
    @Override
    public void calInterest() {
        double interest = bal * interestRate / 100;
        System.out.println("Interest Earned : ■" + interest);
    }
}

class CurrentAcc extends Account {
    double overdraftLimit;
    public CurrentAcc(int accNo, String name, double bal, double overdraftLimit) {
        super(accNo, name, bal, "Current");
        this.overdraftLimit = overdraftLimit;
    }
    @Override
    public void withdraw(double amount) {
        if (bal + overdraftLimit >= amount) {
            bal -= amount;
            System.out.println("■" + amount + " withdrawn successfully.");
        } else {
            System.out.println("Overdraft Limit Exceeded!");
        }
    }
    public void showOverdraftLimit() {
        System.out.println("Overdraft Limit : ■" + overdraftLimit);
    }
}

public class Bank {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        SavingsAcc customer1 = new SavingsAcc(1001, "Bunny", 10000, 5);
        CurrentAcc customer2 = new CurrentAcc(2001, "Arya", 5000, 3000);
        int choice;
        do {
            System.out.println("\n===== BANK MENU =====");
            System.out.println("1. Deposit");
            System.out.println("2. Withdraw");
            System.out.println("3. Transfer");
            System.out.println("4. Calculate Interest");
            System.out.println("5. Display Accounts");
            System.out.println("6. Exit");
            System.out.print("Enter your choice: ");
            choice = sc.nextInt();
            switch (choice) {
                case 1:
                    System.out.println("Select Account:");
                    System.out.println("1. Bunny (Savings)");
                    System.out.println("2. Arya (Current)");
                    int dAcc = sc.nextInt();
                    System.out.print("Enter amount: ");
                    double dAmt = sc.nextDouble();
                    if (dAcc == 1) customer1.deposit(dAmt);
                    else if (dAcc == 2) customer2.deposit(dAmt);
                    else System.out.println("Invalid Account!");
                    break;
                case 2:
                    System.out.println("Select Account:");
                    System.out.println("1. Bunny (Savings)");
                    System.out.println("2. Arya (Current)");
                    int wAcc = sc.nextInt();
                    System.out.print("Enter amount: ");
                    double wAmt = sc.nextDouble();
                    if (wAcc == 1) customer1.withdraw(wAmt);
                    else if (wAcc == 2) customer2.withdraw(wAmt);
                    else System.out.println("Invalid Account!");
                    break;
                case 3:
                    System.out.println("Transfer:");
                    System.out.println("1. Bunny -> Arya");
                    System.out.println("2. Arya -> Bunny");
                    int t = sc.nextInt();
                    System.out.print("Enter amount: ");
                    double tAmt = sc.nextDouble();
                    if (t == 1) customer1.transfer(customer2, tAmt);

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        else if (t == 2) customer2.transfer(customer1, tAmt);
        else System.out.println("Invalid Choice!");
        break;
    case 4:
        System.out.println("1. Bunny (Savings)");
        System.out.println("2. Arya (Current)");
        int i = sc.nextInt();
        if (i == 1) customer1.calInterest();
        else if (i == 2) customer2.calInterest();
        else System.out.println("Invalid Choice!");
        break;
    case 5:
        customer1.displayAccount();
        customer2.displayAccount();
        customer2.showOverdraftLimit();
        break;
    case 6:
        System.out.println("Thank you for using Bank Management System.");
        break;
    default:
        System.out.println("Invalid Choice!");
    }
} while (choice != 6);
}
}

```

Output

```

===== BANK MENU =====
1. Deposit
2. Withdraw
3. Transfer
4. Calculate Interest
5. Display Accounts
6. Exit
Enter your choice: 5

-----
Account Number : 1001
Account Holder : Bunny
Account Type   : Savings
Balance        : ■10000.0

-----
Account Number : 2001
Account Holder : Arya
Account Type   : Current
Balance        : ■5000.0
Overdraft Limit : ■3000.0

```